SOFTWARE ENGINEERING LAB 5

Static Code Analysis
Warehouse Stock Manager

Project: SETAPESU26/L5_54_warehouse-stock

Tools: Pylint • Bandit • Flake8
Python 3.13.1 • Windows CMD

NAME: SHREYASH M YADWAD

SRN: PES2UG24CS493

SEC: H

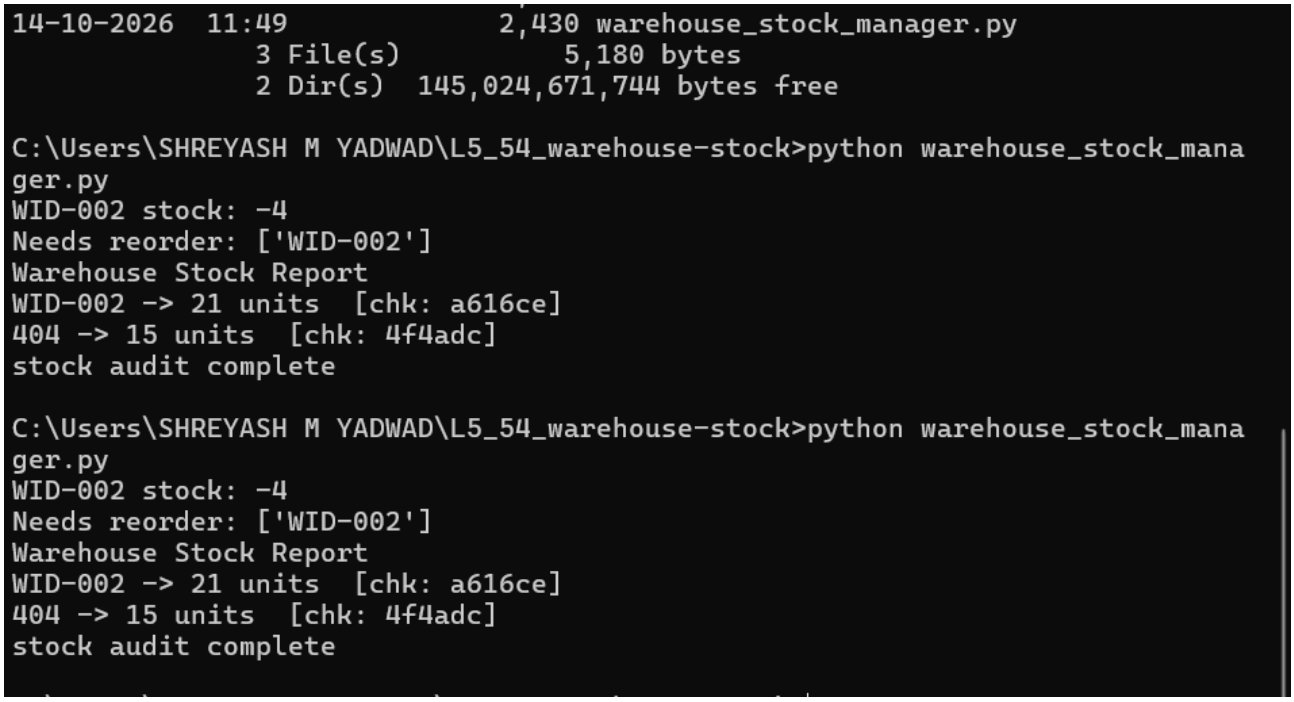
1. Objective and Deliverables

The objective of this lab is to use static-analysis tools to identify and correct code-quality, security, reliability, and style problems in warehouse_stock_manager.py. The lab handout requires a cleaned Python file, an issue table, reflection answers, and before/after reports for Pylint, Bandit, and Flake8. It also specifies three commits for the fixes.

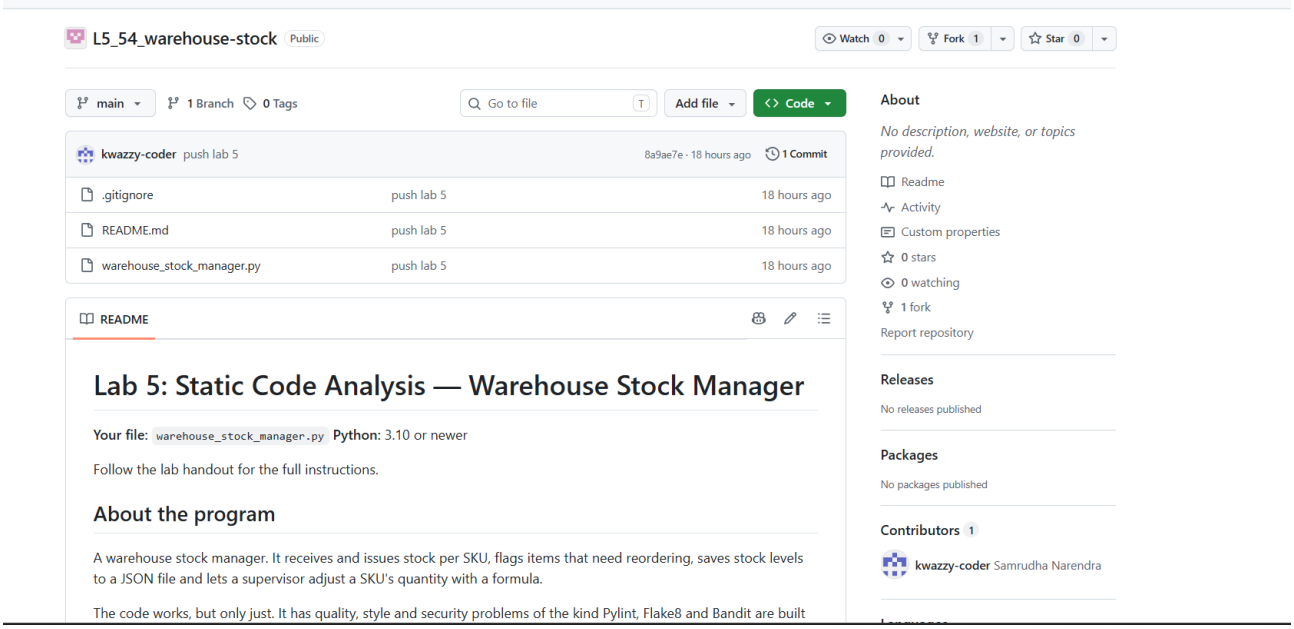
Deliverable	What is included / current evidence
warehouse_stock_manager.py	Corrected version edited during the lab; syntax compilation succeeded.
issues.md	Issue table supplied below for use in the repository.
reports/before/	Original Pylint, Bandit, and Flake8 reports.
reports/after/	Final Pylint, Bandit, and Flake8 reports.
reflection.md	The supplied PDF references four reflection questions but does not display their exact wording; see Section 7.
Git history	First commit confirmed: 74aeb50 — Fix security issues identified by Bandit. A later code-quality commit was planned; verify final history before submission.

2. Original Program Behaviour

The original demo ran and printed stock results, but the output revealed logical problems: WID-002 became -4 units, an integer SKU (404) was accepted, and an excessive removal silently removed WID-001. The original code also used eval() for formula handling and for a print-only statement.



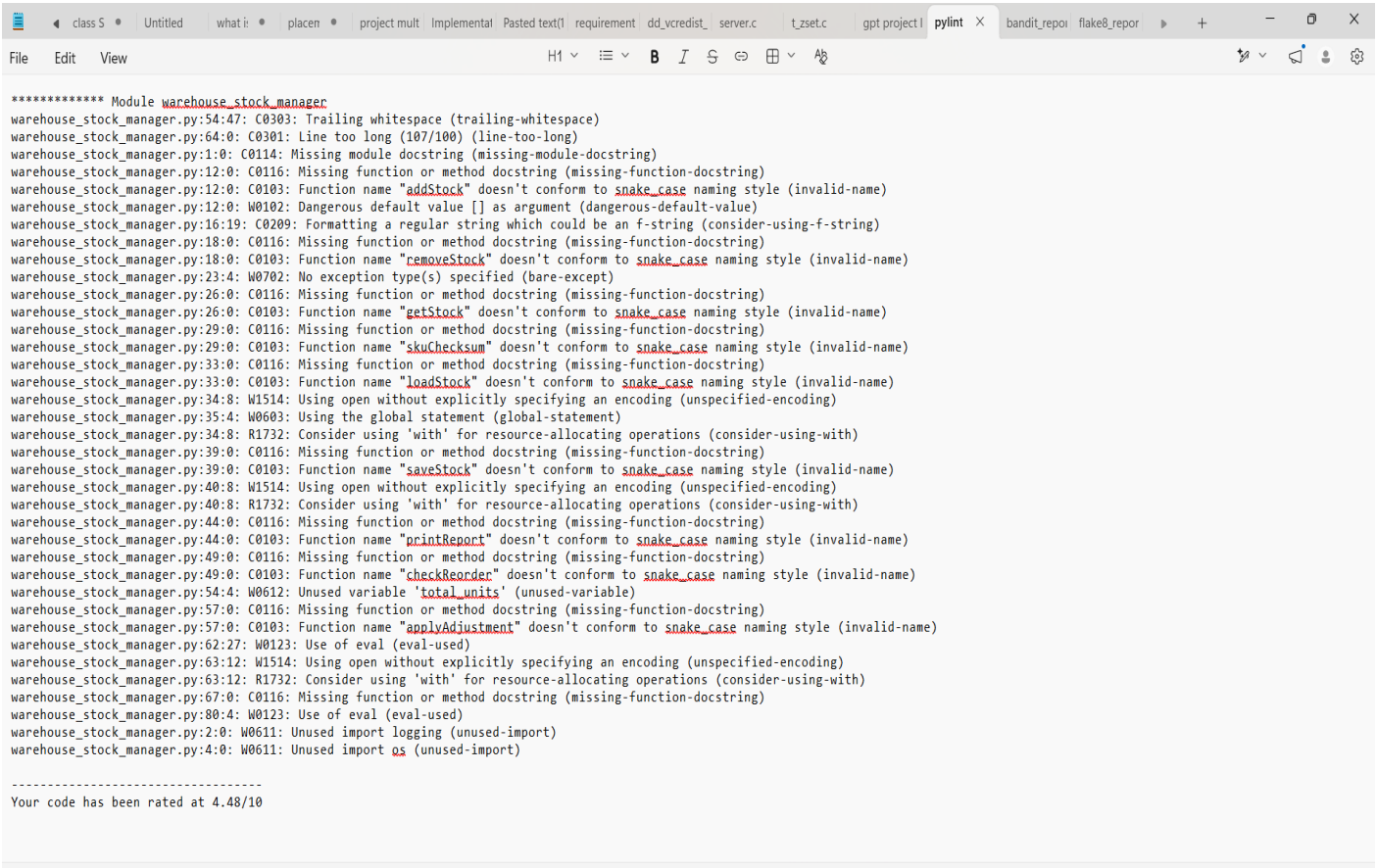
Screenshot 1. Original program output showing negative stock and an integer SKU.



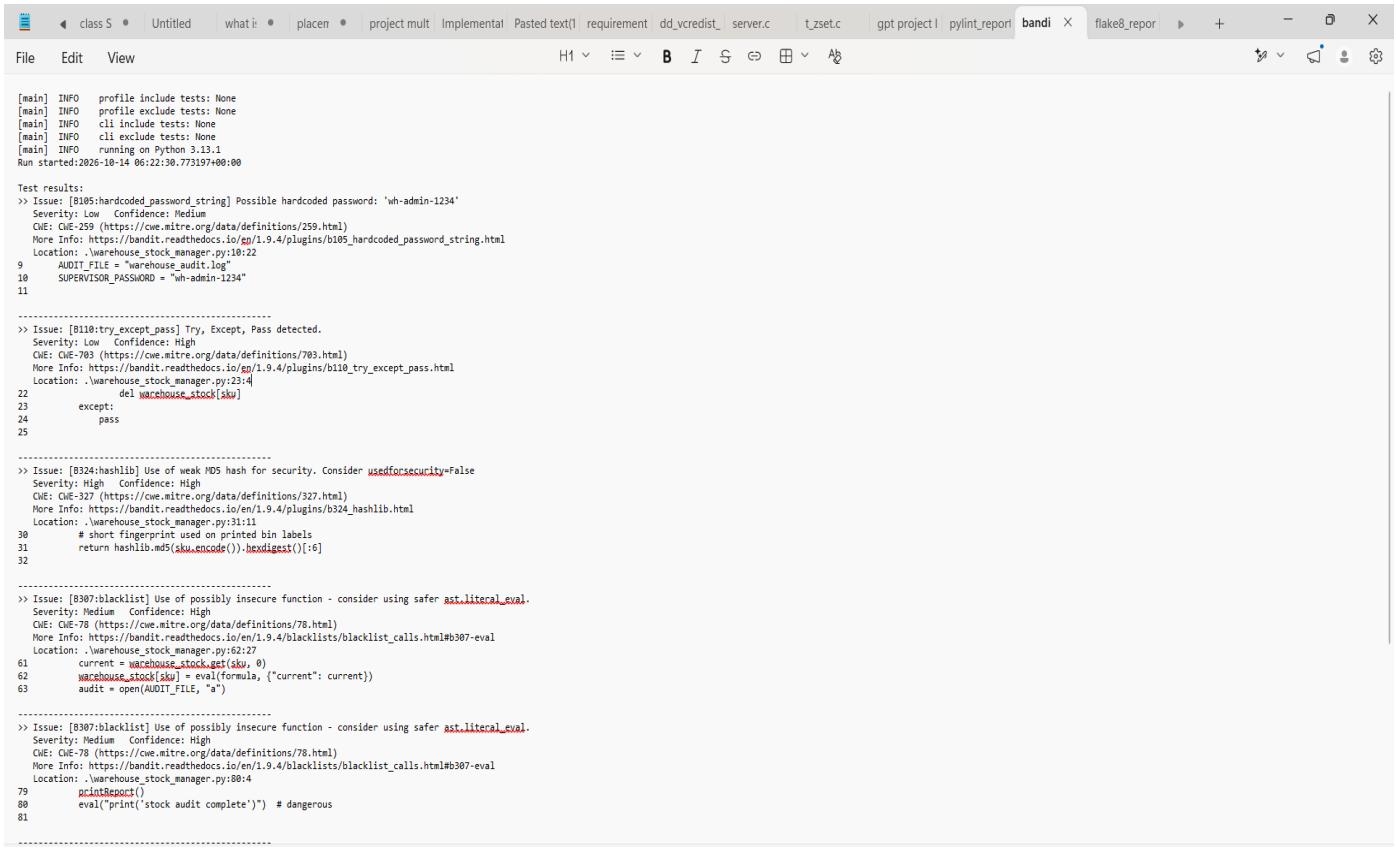
Screenshot 2. Assigned GitHub repository and README.

3. Before Corrections: Static Analysis

The original reports were captured before changes. Pylint rated the original code 4.48/10. Bandit reported five findings in the screenshot, including a hardcoded password, a bare exception, MD5 usage, and two eval() calls. Flake8 reported unused imports, formatting violations, a bare exception, an unused variable, and other style issues.



Screenshot 3. Original Pylint report with score 4.48/10.



```
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.1
Run started:2026-10-14 06:22:30.773197+00:00

Test results:
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'wh-admin-1234'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\warehouse_stock_manager.py:10:22
9       AUDIT_FILE = "warehouse_audit.log"
10      SUPERVISOR_PASSWORD = "wh-admin-1234"
11

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\warehouse_stock_manager.py:23:4
22                  del warehouse_stock[sku]
23          except:
24              pass
25

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\warehouse_stock_manager.py:31:11
30          # short fingerprint used on printed bin labels
31          return hashlib.md5(sku.encode()).hexdigest()[:6]
32

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\warehouse_stock_manager.py:62:27
61          current = warehouse_stock.get(sku, 0)
62          warehouse_stock[sku] = eval(formula, {"current": current})
63          audit = open(AUDIT_FILE, "a")

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\warehouse_stock_manager.py:80:4
79          printReport()
80          eval("print('stock audit complete')")  # dangerous
81

--------------------------------------------------
```

Screenshot 4. Original Bandit report: hardcoded password and bare exception findings.

```
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\warehouse_stock_manager.py:31:11
30          # short fingerprint used on printed bin labels
31          return hashlib.md5(sku.encode()).hexdigest()[:6]
32

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\warehouse_stock_manager.py:62:27
61          current = warehouse_stock.get(sku, 0)
62          warehouse_stock[sku] = eval(formula, {"current": current})
63          audit = open(AUDIT_FILE, "a")

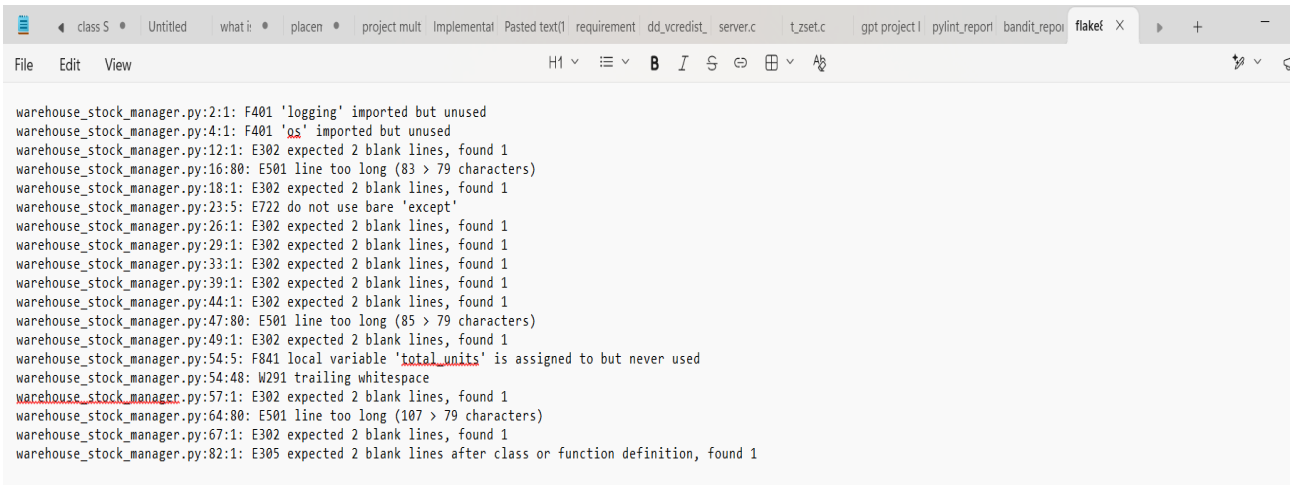
--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\warehouse_stock_manager.py:80:4
79          printReport()
80          eval("print('stock audit complete')")  # dangerous
81

--------------------------------------------------

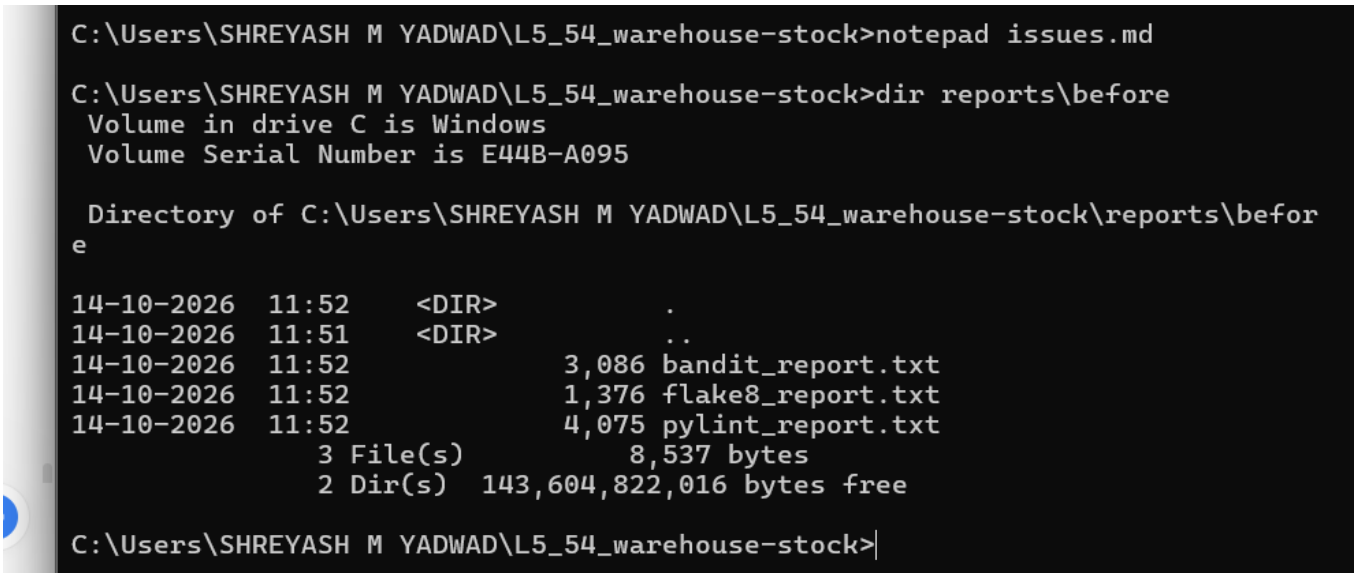
Code scanned:
        Total lines of code: 67
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 2
                High: 1
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 4
Files skipped (0):
```

Screenshot 5. Original Bandit report: MD5 and eval() findings, with summary counts.



Screenshot 6. Original Flake8 report showing formatting and code-quality findings.



Screenshot 7. Original reports saved in reports/before/.

4. issues.md — Final Issue Table

Copy this table into issues.md. Line numbers refer to the original source/report screenshots and may shift after code changes; verify them against the original committed file if your instructor checks exact lines.

Issue	Type	Original line(s)	Description	Fix approach
Hardcoded supervisor password	Security	10	Password was embedded directly in source code.	Read the actual secret from an environment variable.
Unsafe eval() calls	Security	62, 80	Expressions were executed as Python code, including a call used only to print a message.	Replace formula evaluation with a restricted AST arithmetic evaluator; use print() for the message.
Bare exception handling	Reliability / Security	22–24	except: pass silently hid errors.	Validate stock operations and report insufficient stock explicitly.
Weak MD5 checksum	Security	31	Bandit flagged MD5 as a weak hash.	Use SHA-256 for the non-security bin-label fingerprint.
Mutable default argument	Bug / Code quality	12	history=[] could be shared between calls.	Use history=None and initialize a new list inside the function.

Missing SKU and quantity validation	Input validation	12, 18, 72–78	The original demo accepted integer SKUs and negative stock quantities.	Require a non-empty string SKU and non-negative integer quantity.
Missing-SKU lookup failure	Reliability	26	warehouse_stock[sku] raised KeyError for an absent SKU.	Use warehouse_stock.get(sku, 0).
Manual file management	Resource management	34–41, 63–64	Files were opened and closed manually without explicit encoding.	Use with open(..., encoding='utf-8') context managers.
Unused calculation	Maintainability	54	total_units was calculated but never used.	Remove the unused variable/calculation.
Naming and documentation	Style / Maintainability	Multiple	CamelCase names and missing docstrings lowered code quality.	Use snake_case function names and add module/function docstrings.
Formatting violations	Style	Multiple	Flake8 reported long lines, whitespace and blank-line violations.	Apply the reported PEP 8 corrections.

5. After Corrections: Changes Made

The principal before/after changes are summarized below.

Before	After	Reason
eval(formula, {'current': current})	evaluate_formula(formula, current) using a restricted AST	Avoid arbitrary code execution while supporting approved arithmetic.
eval("print('stock audit complete')")	print("Stock audit complete")	A string should not be executed as code just to print a message.
SUPERVISOR_PASSWORD = "wh-admin-1234"	Read the secret using os.environ.get(...)	Avoid storing the actual password in source code.
history=[]	history=None, then initialize a new list	Prevent shared mutable defaults.
except: pass	Validate inputs and return/report insufficient stock	Do not silently hide failures.
hashlib.md5(...)	hashlib.sha256(...)	Address Bandit's weak-hash warning for this fingerprint.
open()/close() calls	with open(..., encoding='utf-8')	Ensure file resources are closed and encoding is explicit.
camelCase names / no docstrings	snake_case names and docstrings	Improve readability and Pylint compliance.
Unused total_units and formatting issues	Remove unused calculation and apply formatting fixes	Reduce lint warnings.

```
    f.write(json.dumps(warehouse_stock))
    f.close()

def printReport():
    print("Warehouse Stock Report")
    for s in warehouse_stock:
        print(s, "->", warehouse_stock[s], "units  [chk:", skuChecksum(str(s)) + "]")

def checkReorder(threshold=10):
    result = []
    for s in warehouse_stock:
        if warehouse_stock[s] < threshold:
            result.append(s)
    total_units = sum(warehouse_stock.values())
    return result

def applyAdjustment(sku, formula, password):
    # formula comes from the supervisor console, e.g. "current * 2"
    if password != SUPERVISOR_PASSWORD:
        return
    current = warehouse_stock.get(sku, 0)
    warehouse_stock[sku] = eval(formula, {"current": current})
    audit = open(AUDIT_FILE, "a")
    audit.write("adjusted " + str(sku) + " using formula " + formula + " at " +
str(datetime.now()) + "\n")
    audit.close()

def main():
    addStock("WID-001", 40)
    addStock("WID-002", -4)
    addStock(404, 15)  # invalid SKU type, no check
    removeStock("WID-001", 12)
    removeStock("GHOST-9", 1)
    removeStock("WID-001", 500)
    print("WID-002 stock:", getStock("WID-002"))
    print("Needs reorder:", checkReorder())
    applyAdjustment("WID-002", "current + 25", "wh-admin-1234")
    saveStock()
    loadStock()
    printReport()
    print("stock audit complete")

main()
```

Screenshot 8. Code editor during security corrections.

```
import ast
import operator
import os
import json
import logging
import hashlib
import os
from datetime import datetime

# Global variables
warehouse_stock = {}
AUDIT_FILE = "warehouse_audit.log"
SUPERVISOR_PASSWORD = os.environ.get("WAREHOUSE_SUPERVISOR_PASSWORD")
def addStock(sku="default", qty=0, history=[]):
    if not sku:
        return
    warehouse_stock[sku] = warehouse_stock.get(sku, 0) + qty
    history.append("%s: Received %d units of %s" % (str(datetime.now()), qty, sku))

def removeStock(sku, qty):
    try:
        warehouse_stock[sku] -= qty
        if warehouse_stock[sku] <= 0:
            del warehouse_stock[sku]
    except:
        pass

def getStock(sku):
    return warehouse_stock[sku]

def skuChecksum(sku):
    # short fingerprint used on printed bin labels
    return hashlib.md5(sku.encode()).hexdigest()[:6]

def loadStock(file="warehouse.json"):
    f = open(file, "r")
    global warehouse_stock
    warehouse_stock = json.loads(f.read())
    f.close()

def saveStock(file="warehouse.json"):
    f = open(file, "w")
    f.write(json.dumps(warehouse_stock))
```

Ln 12, Col 35 | 3,374 characters | Plain text | 90% | Windows (CRLF) | UTF-8

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Screenshot 9. Imports and original functions before the full cleanup.

```
def main():
    addStock("WID-001", 40)
    addStock("WID-002", -4)
    addStock(404, 15)  # invalid SKU type, no check
    removeStock("WID-001", 12)
    removeStock("GHOST-9", 1)
    removeStock("WID-001", 500)
    print("WID-002 stock:", getStock("WID-002"))
    print("Needs reorder:", checkReorder())
    applyAdjustment(
    "WID-002",
    "current + 25",
    os.environ.get("WAREHOUSE_SUPERVISOR_PASSWORD", ""),
)
    saveStock()
    loadStock()
    printReport()
    print("stock audit complete")
```

Screenshot 10. Updated main() call using the environment variable.

```
def saveStock(file="warehouse.json"):
    f = open(file, "w")
    f.write(json.dumps(warehouse_stock))
    f.close()

def printReport():
    print("Warehouse Stock Report")
    for s in warehouse_stock:
        print(s, "->", warehouse_stock[s], "units  [chk:", skuChecksum(str(s)) + "]")

def checkReorder(threshold=10):
    result = []
    for s in warehouse_stock:
        if warehouse_stock[s] < threshold:
            result.append(s)
    total_units = sum(warehouse_stock.values())
    return result

def evaluate_formula(formula, current):
    """Safely evaluate basic arithmetic using the current stock value."""
    allowed_operators = {
        ast.Add: operator.add,
        ast.Sub: operator.sub,
        ast.Mult: operator.mul,
        ast.Div: operator.truediv,
    }

    tree = ast.parse(formula, mode="eval")

    def evaluate(node):
        if isinstance(node, ast.Expression):
            return evaluate(node.body)

        if isinstance(node, ast.Constant) and type(node.value) in (int, float):
            return node.value

        if isinstance(node, ast.Name) and node.id == "current":
            return current

        if isinstance(node, ast.BinOp):
            operation = allowed_operators.get(type(node.op))
            if operation is None:
```

Screenshot 11. Restricted evaluate_formula() helper.

```
    total_units = sum(warehouse_stock.values())
    return result

def evaluate_formula(formula, current):
    """Safely evaluate basic arithmetic using the current stock value."""
    allowed_operators = {
        ast.Add: operator.add,
        ast.Sub: operator.sub,
        ast.Mult: operator.mul,
        ast.Div: operator.truediv,
    }

    tree = ast.parse(formula, mode="eval")

    def evaluate(node):
        if isinstance(node, ast.Expression):
            return evaluate(node.body)

        if isinstance(node, ast.Constant) and type(node.value) in (int, float):
            return node.value

        if isinstance(node, ast.Name) and node.id == "current":
            return current

        if isinstance(node, ast.BinOp):
            operation = allowed_operators.get(type(node.op))
            if operation is None:
                raise ValueError("Unsupported operator")
            return operation(evaluate(node.left), evaluate(node.right))

        raise ValueError("Invalid adjustment formula")

    return evaluate(tree)

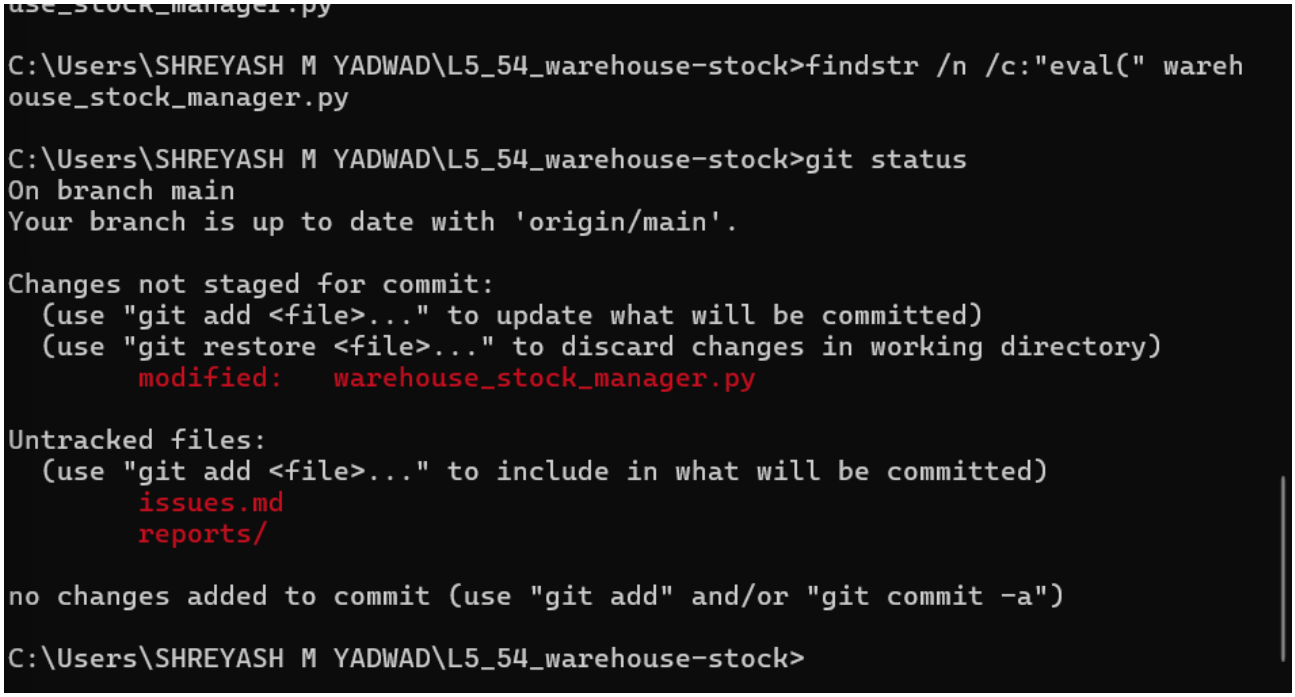
def applyAdjustment(sku, formula, password):
    # formula comes from the supervisor console, e.g. "current * 2"
    if password != SUPERVISOR_PASSWORD:
        return
```

Screenshot 12. End of formula helper and start of apply_adjustment().

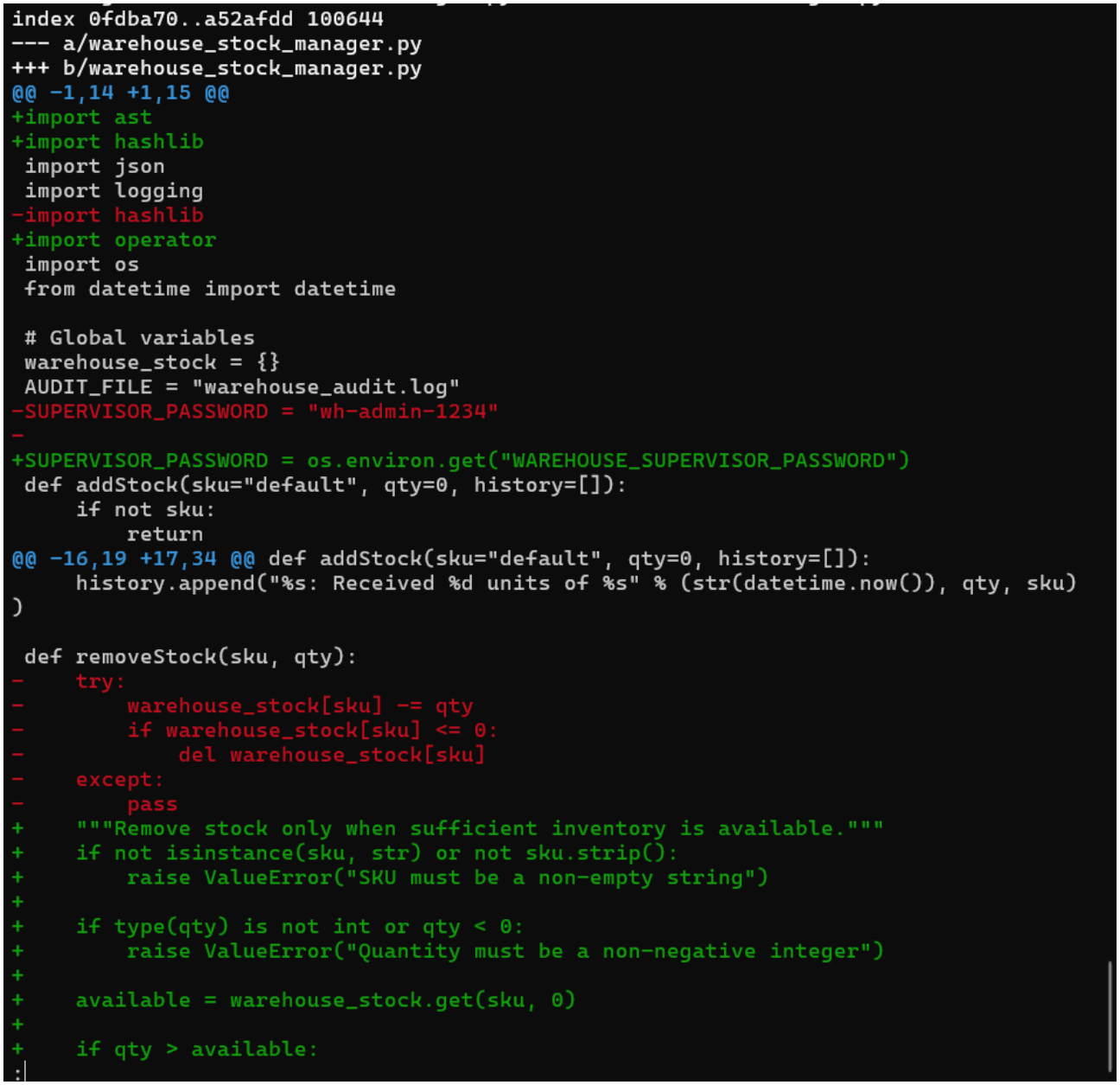
```
C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m py_compile wareho
use_stock_manager.py

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>
```

Screenshot 13. py_compile completed without syntax errors.



Screenshot 14. No literal eval(match and Git status showing changed source and untracked evidence.



```
index 0fdba70..a52afdd 100644
--- a/warehouse_stock_manager.py
+++ b/warehouse_stock_manager.py
@@ -1,14 +1,15 @@
+import ast
+import hashlib
 import json
 import logging
-import hashlib
+import operator
 import os
 from datetime import datetime

 # Global variables
 warehouse_stock = {}
 AUDIT_FILE = "warehouse_audit.log"
-SUPERVISOR_PASSWORD = "wh-admin-1234"
-
+SUPERVISOR_PASSWORD = os.environ.get("WAREHOUSE_SUPERVISOR_PASSWORD")
 def addStock(sku="default", qty=0, history=[]):
     if not sku:
         return
@@ -16,19 +17,34 @@ def addStock(sku="default", qty=0, history=[]):
     history.append("%s: Received %d units of %s" % (str(datetime.now()), qty, sku)
)

 def removeStock(sku, qty):
-    try:
-        warehouse_stock[sku] -= qty
-        if warehouse_stock[sku] <= 0:
-            del warehouse_stock[sku]
-    except:
-        pass
+    """Remove stock only when sufficient inventory is available."""
+    if not isinstance(sku, str) or not sku.strip():
+        raise ValueError("SKU must be a non-empty string")
+
+    if type(qty) is not int or qty < 0:
+        raise ValueError("Quantity must be a non-negative integer")
+
+    available = warehouse_stock.get(sku, 0)
+
+    if qty > available:
:
```

Screenshot 15. Git diff showing security changes to the source.

6. After Corrections: Verification Results

The latest screenshots show a Pylint score of 10.00/10, Flake8 producing no output, and Bandit reporting "No issues identified." These results apply to the version scanned in those screenshots. Rerun the tools after any further edit and keep the reports in reports/after/ synchronized with the final file.

Tool / check	Before	After shown in evidence
Pylint	4.48/10	10.00/10
Bandit	5 findings in the original report	No issues identified
Flake8	Multiple formatting/style findings	No output from the final command
Python syntax	Original program ran, but had logical issues	py_compile completed without error

```
ager.py

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m py_compile wareho
use_stock_manager.py

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m bandit -r warehou
se_stock_manager.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.1
Run started:2026-10-14 06:50:44.196471+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 117
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>
```

Screenshot 16. Bandit scan after the first set of security changes; this earlier run reported no issues.

```
C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>git add warehouse_stock_manager.py

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>git commit -m "Fix security issues identified by Bandit"
[main 74aeb50] Fix security issues identified by Bandit
 1 file changed, 81 insertions(+), 13 deletions(-)

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>git log -1 --oneline
74aeb50 (HEAD -> main) Fix security issues identified by Bandit

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        issues.md
        reports/

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>
```

Screenshot 17. First Git commit confirmed: 74aeb50 — Fix security issues identified by Bandit.

```
ing-function-docstring)
warehouse_stock_manager.py:120:0: C0103: Function name "applyAdjustment" doesn't co
nform to snake_case naming style (invalid-name)
warehouse_stock_manager.py:126:12: W1514: Using open without explicitly specifying
an encoding (unspecified-encoding)
warehouse_stock_manager.py:126:12: R1732: Consider using 'with' for resource-alloca
ting operations (consider-using-with)
warehouse_stock_manager.py:130:0: C0116: Missing function or method docstring (miss
ing-function-docstring)

--------------------------------------------------------------------
Your code has been rated at 6.80/10 (previous run: 4.48/10, +2.32)


C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>notepad warehouse_stock_manager.py


C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m py_compile warehouse_sto
ck_manager.py

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m pylint warehouse_stock_m
anager.py

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 6.80/10, +3.20)


C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>
```

Screenshot 18. Pylint score improved from 4.48/10 to 10.00/10.

```
C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m flake8 warehouse_stock_manager.py
warehouse_stock_manager.py:200:15: E201 whitespace after '('

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m bandit -r warehouse_stock_manager.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.1
Run started:2026-10-14 06:58:28.768679+00:00

Test results:
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'WAREHOUSE_SUPERVISOR_PASSWORD'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\warehouse_stock_manager.py:16:15
15      STOCK_FILE = "warehouse.json"
16      PASSWORD_ENV = "WAREHOUSE_SUPERVISOR_PASSWORD"
17

--------------------------------------------------

Code scanned:
        Total lines of code: 166
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 0
Files skipped (0):

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m pylint warehouse_stock_manager.py > reports\after\pylint_report.txt 2>&1

C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>
```

Screenshot 19. Final Flake8 command with no output and an intermediate Bandit warning about the environment-variable name.

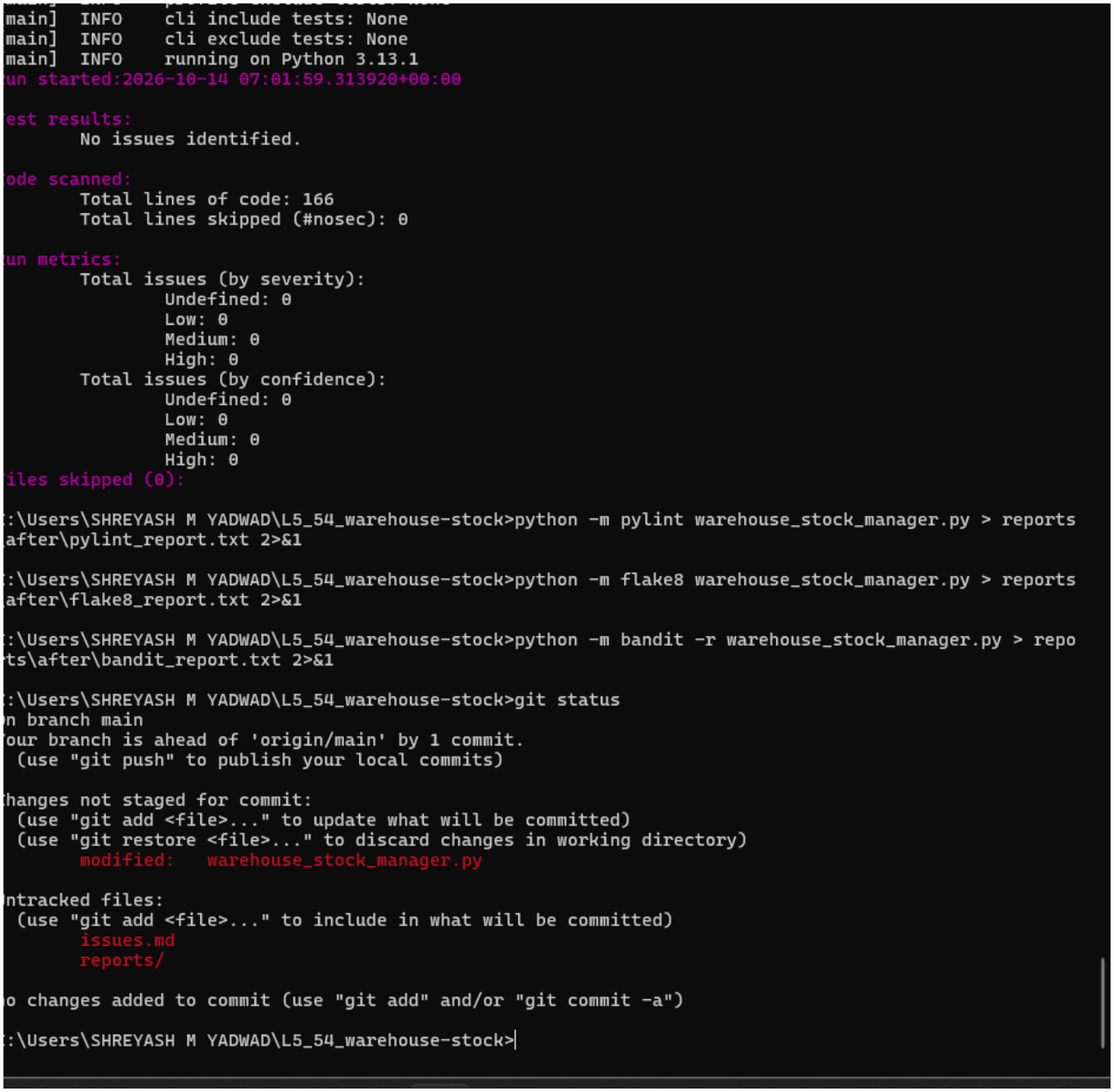
```
C:\Users\SHREYASH M YADWAD\L5_54_warehouse-stock>python -m bandit -r warehouse_stock_manager.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.1
Run started:2026-10-14 07:01:59.313920+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 166
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Screenshot 20. Final Bandit output: no issues identified.



Screenshot 21. Final report commands and Git status.

7. Reflection

Note: The supplied three-page student handout references reflection questions, but the provided PDF does not include the exact four question statements. The answers below are draft reflections based on the handout's learning outcomes. If the instructor supplied a separate question sheet, replace the headings with the exact questions.

1. Purpose of static code analysis

Static analysis examines source code without needing to execute every possible path. In this lab, Pylint helped identify maintainability and code-quality issues, Flake8 identified style violations, and Bandit highlighted security-sensitive patterns.

2. Most important issues found

The most important findings were the unsafe eval() calls, the hardcoded supervisor password, broad exception handling, and the weak MD5 fingerprint. The original demo also exposed behavioural problems such as negative stock and accepting an integer SKU.

3. How the fixes were verified

The formula evaluation was restricted to approved arithmetic, the password was read from an environment variable, stock inputs were validated, and file operations were updated. The revised file compiled successfully; the final screenshots show Pylint at 10.00/10, Flake8 with no output, and Bandit with no issues identified.

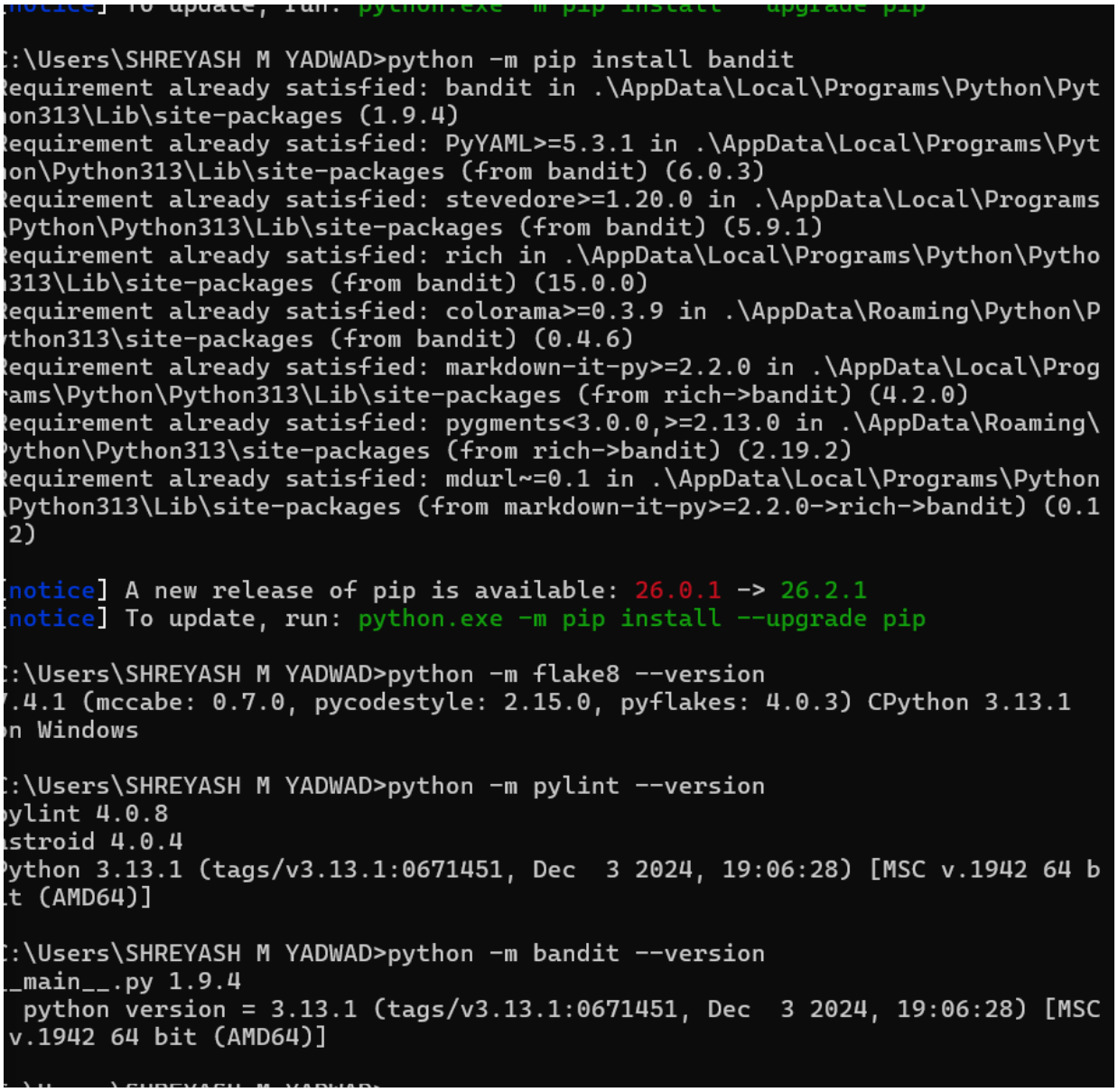
4. Contribution to maintainability

Static analysis gives repeatable feedback before submission or release. It helps catch defects early, encourages consistent naming and formatting, and highlights security risks that might otherwise be missed during manual review. Tool output should still be combined with functional tests and code review.

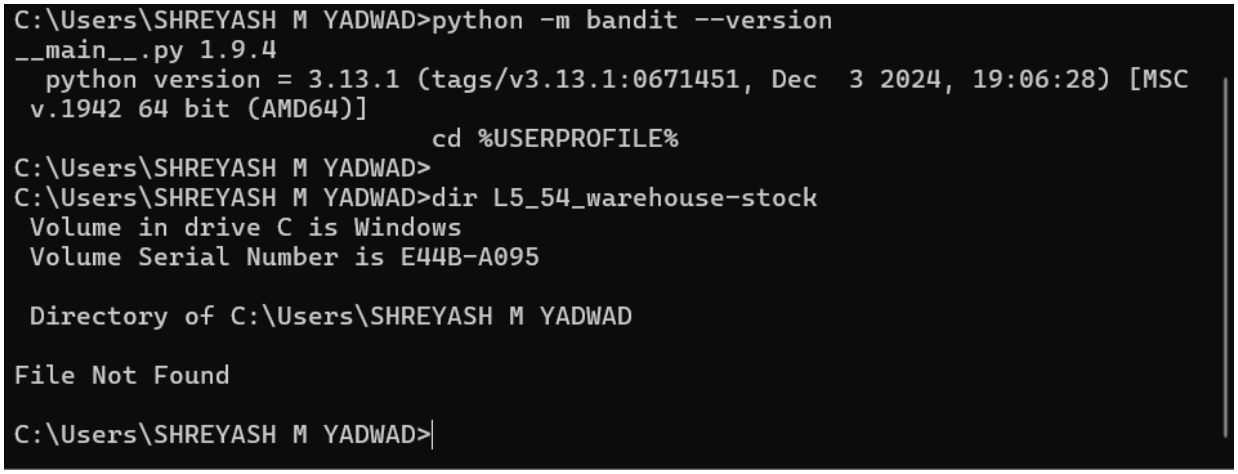
8. Git Commit and Submission Checklist

Requirement	Evidence / action
Three commits	The first commit is confirmed as 74aeb50. Verify that there are three distinct commits for Bandit, Pylint, and Flake8-related changes as the handout requests.
Corrected source	Commit the final warehouse_stock_manager.py after verifying the tests and reports.
Issue table	Save the issue table from Section 4 as issues.md in the repository.
Before/after reports	Keep all six report files under reports/before/ and reports/after/.
Reflection	Save reflection.md and ensure it answers the instructor's exact questions if a separate question sheet exists.
Push to GitHub	Run git status, git log --oneline -3, then git push origin main when the intended commits and files are ready.

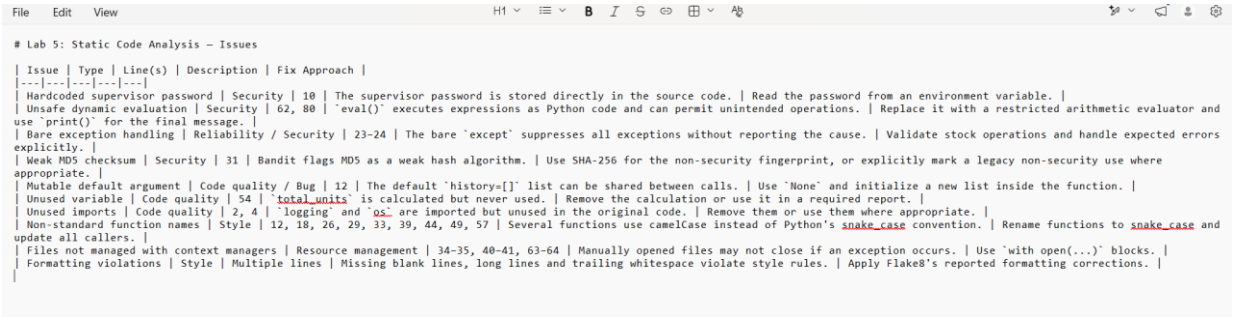
Appendix A. Additional Screenshots



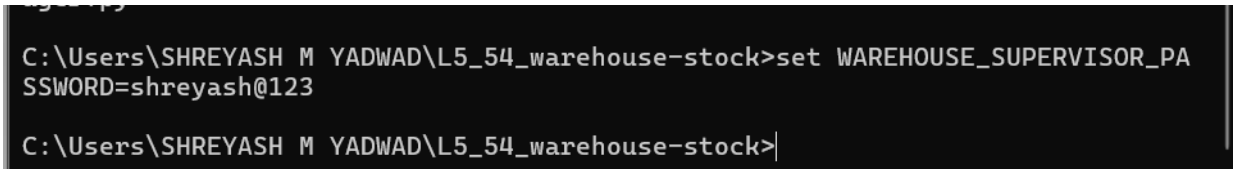
Additional screenshot evidence: image(20261014-042457).png



Additional evidence: initial repository-folder check before cloning.



Additional evidence: issues.md draft displayed in the editor.



Additional evidence: environment variable configured in CMD and code edit in progress.